Jose Alberto Tupayachi Silva

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Summary

Industrial & Systems Engineer specialized in data science and process automation with strong problem-solving and teamwork skills. Experienced in developing end-to-end AI solutions, working in Agile sprints. Published research includes npj Digital Medicine, Computers & Electrical Engineering and Computers & Industrial Engineering.

Education

University of Tennessee, Knoxville
Ph.D., Industrial Engineering May 2024 - May 2027

University of Tennessee, Knoxville
M.S., Industrial Engineering Aug 2022 - May 2024

Pontifical Catholic University of Peru
B.S., Industrial Engineering Mar 2014 - May 2019

Experience

Oak Ridge National Laboratory **Mar 2025 - Present**
Developer *Oak Ridge, TN*

- Deploy a pipeline for spectral (PCA - cuML) and spatial (alpha-shape) ToF-SIMS - 6PPD Analysis based on .itax/itmx IONTOF .
- Fine-tune a vision (Swin Transformer) to quality screen wind speed and direction from weather stations used in atmospheric plume modeling.
- Develop an explainable ML model for atmospheric and geospatial datasets (DL and 1DCNN).
- Automate geospatial dashboard generator with natural language knowledge graph reasoning (llama.cpp, langchain, vLLM).
- Deploy retrieval augmented generation for 1000+ lithium breeder blanket papers with research direction identification (ontology).

University of Tennessee, Knoxville **Aug 2022 - Present**
Graduate Research Assistant *Knoxville, TN*

- Deployed an AI-powered digital twin for freight transportation with linear optimization, real-time routing, and SLURM parallel processing.
- Coded a mobile health applications with Android, iOS, watchOS, and Wear OS distributions (Flutter).
- Integrated an Intermodal routing optimizer with ontology-guided REST API integration.
- Developed a Traffic-Weather Graph Convolutional Network for EV charging demand forecasting that feeds Traffic Message Channels into a ensemble architecture.

Indra **Mar 2022 - Aug 2022**
Data Engineer

- Built and maintained scalable ETL/ELT pipelines using Python, Scala, and Shell scripting on UNIX/Linux, processing high-volume datasets across Agile delivery cycles to support downstream analytics and reporting.
- Engineered distributed data processing systems using Scala/PySpark on Apache Spark, Hadoop, and Hive/HQL, and led migration of a legacy Oracle warehouse to a partitioned AWS-based cloud lakehouse (S3, EMR), optimizing Spark memory and resource allocation to improve job execution efficiency.
- Designed data models to support data warehousing, and implemented entity-resolution and deduplication logic to unify and standardize records across varied sources.
- Automated CI/CD deployment pipelines using Jenkins, streamlining builds, unit testing, and application deployment including live deployments, rollback procedures, and backup strategies to reduce manual effort and deployment risk.

Globokas **Jan 2022 - Mar 2022**

Business Intelligence Analyst

- Developed dashboards and KPI reporting systems supporting strategic and operational decision-making for multiple business clients.
- Integrated heterogeneous data sources (REST APIs, internal systems, external platforms) into unified analytics pipelines.

Enel **Nov 2020 - Dec 2021**

Data Analyst

- Built interactive Power BI and Tableau dashboards for operational monitoring and KPI reporting.
- Managed SQL/T-SQL databases and Salesforce integrations via API queries.
- Developed a PyQt desktop application to automate invoice verification and special-case processing.
- Engineered customer segmentation models using unsupervised machine learning (K-Means and hierarchical clustering) on payment history and demographics, enabling collection teams to prioritize outreach and improve payment-recovery.

GPS Management **Aug 2020 - Nov 2020**

Developer - Portfolio Valuation

- Implemented machine learning models for non-performing loan (NPL) recovery and portfolio valuation.
- Automated scoring workflows, reducing manual processing time.

Quiron Salud **Jan 2019 - Aug 2019**

IT Intern

- Developed and maintained healthcare plans information systems.
- Supported electronic health record integration and database administration for clinical operations.

Selected Publications

- Tupayachi, J., Khan, A. N., & Li, X. (2026). Scalable decentralized prognostics for industrial systems under data heterogeneity. *Computers & Electrical Engineering*, 133, Article 111023. https://doi.org/10.1016/j.compeleceng.2026.111023
- Khan, A. N., Tupayachi, J., Dabbs, W., Chamberlin, S., Yao, B., & Li, X. (2026). CliniSense AI for automated clinical skills assessment with real-time feedback. *npj Digital Medicine*. https://doi.org/10.1038/s41746-026-03100-8
- Wyatt, T. H., Lowe, S., Tupayachi, J., & Li, X. (2026). Design and usability testing of SmartSHOTS: A mobile app to reduce vaccine barriers for children 0–24 months. *Health Informatics Journal*, 32(1).https://doi.org/10.1177/14604582261419334
- Xu, H., Sun, Y., Tupayachi, J., Omitaomu, O., Zlatanova, S., & Li, X. (2026). Towards autonomous freight intermodal optimization via generative AI and agentic digital twins. *Computers & Industrial Engineering*, Article 112464.https://doi.org/10.1016/j.cie.2026.112464
- Xu, H., Li, X., Tupayachi, J., Lian, J. J., & Omitaomu, O. A. (2024). Automating bibliometric analysis with sentence transformers and RAG. *Proceedings of the 2nd ACM SIGSPATIAL International Workshop on Advances in Urban-AI*, 1–10. https://doi.org/10.1145/3681780.3697252
- Tupayachi, J., Ferguson, M. M., & Li, X. (2024). A simulation-based real-time deep reinforcement learning approach for fighting wildfires. *Proceedings of the 2024 Annual Modeling and Simulation Conference (ANNSIM)*, 1–12. https://doi.org/10.23919/annsim61499.2024.10732085
- Tupayachi, J., Xu, H., Omitaomu, O. A., Camur, M. C., Sharmin, A., & Li, X. (2024). Towards next-generation urban decision support systems through AI-powered construction of scientific ontology using large language models. *Smart Cities*, 7(5), 2392–2421. https://doi.org/10.3390/smartcities7050094
- + Publications: https://scholar.google.com/citations?user=Kyn6Et8AAAAJ&hl=en&oi=ao

Skills

AI and Machine Learning

- Machine Learning Engineering, LLMs, RAG, Agentic AI Workflows, CNNs, Vision Transformers, TensorFlow, PyTorch, Statistical Modeling, Predictive Modeling, Clustering, Regression, Tree-Based and Boosting Models (Random Forest, LightGBM, XGBoost), Model Tuning and Calibration, Explainable AI

LLM and Agentic Frameworks

- LangChain, LangGraph, LiteLLM, vLLM, Ollama, QLoRA Finetuning

Cloud and Infrastructure

- AWS (S3, EMR, Redshift, SageMaker, ECS), Microsoft Azure, Google Cloud, Docker, Kubernetes, UNIX/Linux, KVM, Slurm, CI/CD, Jenkins, Git, GitLab

Data Engineering

- Python, SQL, Scala, Apache Spark, Hadoop/MapReduce, Hive, Kafka, ETL/ELT, Batch and Real-Time Streaming, Delta Lake, Databricks Workflows, Unity Catalog, dbt, Airflow, Prefect, Data Modeling, Entity Resolution, Data Quality, Anomaly Detection, Snowflake, Amazon Redshift, Dimensional Modeling, Data Warehouse Architecture, Query and Performance Optimization

Databases

- PostgreSQL, MySQL, SQL Server, MongoDB, Cassandra

Programming Languages

- Java, R, SAS, STATA, Bash, Dart, PHP, C++, CUDA, SPSS

Web, Frameworks, and APIs

- Django, Flask, Streamlit, Dash, React, Flutter, Laravel, REST and GraphQL APIs

Optimization and Simulation

- Gurobi, CPLEX, AnyLogic, NetworkX

Geospatial

- WMS, GeoServer, OverPass API, Turf.js

Certifications

- MS Transact-SQL (T-SQL), Microsoft Fabric DP-600 & DP-700, AWS, GCP

Awards & Honors

- IISE Future Faculty Fellows (2025-2026), USA - Awarded
- Best Paper Award, CIE51 (51st Conference on Computers and Industrial Engineering), Sydney, Australia (2024)
- IISE DAIS Student Mobile App Competition - Winner, SmartShots Project, Montreal (2024)
- HIDA Helmholtz Visiting Researcher, Karlsruhe Institute of Technology (KIT), Germany - Awarded (2024)
- Holiday Fellowship, University of Tennessee, Knoxville (2022, 2023, 2024)

Teaching Experience

Aug 2022 - Dec 2022

University of Tennessee, Knoxville — *Knoxville, TN*

- IE406/408: Simulation - Fall 2022
- IE483: Reliability Engineering - Spring 2023
- IE 533: Theory and Practice of Engineering Management - Fall 2026

Clients & Sponsors

ORNL github.com/jtupayachi/smartgiscavp demo2.recoil.ise.utk.edu **Mar 2025 - Present**

Developer

- Funding: Nuclear Safety Research and Development (NSRD).

RECOIL m1.recoil.ise.utk.edu **Jul 2024 - Mar 2026**

Lead AI/Data Engineer

- Funding: U.S. Department of Energy ARPA-E, Project #DE-AR0001780.

SmartShots iise.org/Details.aspx?id=33697 play.google.com/store/apps/details?id=com.ilab.smartshots **Dec 2023 - Dec 2024**

Full-Stack Developer

- Winner, IISE DAIS Student Mobile App Competition 2024.Funding: Tennessee Department of Health.

Active Caregiver's Toolkit play.google.com/store/apps/details?id=com.ilabutk.fe_rich **Jul 2024 - Present**

Mobile Developer

- Funding: Hillman Emergent Innovation (HEI).

ORNL github.com/jtupayachi/tof-sims-driver **Mar 2025 - Present**

Developer

- Funding: Fusion Energy Sciences (FES)

References

Xueping Li
UTK *Professor*
- xli27@utk.edu

Xiao-Ying Yu
ORNL *Group Leader*
- yuxiaoying@ornl.gov

Bill Hicks
UTK *Instructor*
- william.hicks@tennessee.edu

Carla Arbogast
UTK *Director MSIE*
- carboga1@utk.edu